Chapter 1 Kinematics of a Particle

Exercise 1-4

Answer

Given the acceleration $a(t) = 4 + 3t$. At $t = 0$, $x_0 = 2\ \mathrm{m}$ and $v_0 = 5\ \mathrm{m \cdot s^{-1}}$.

From $a = \frac{\mathrm{d}v}{\mathrm{d}t}$ we get:

$$\mathrm{d}v = a\,\mathrm{d}t = (4 + 3t)\,\mathrm{d}t$$

Integrating both sides:

$$\int_{v_0}^{v} \mathrm{d}v = \int_{0}^{t} (4 + 3t)\,\mathrm{d}t$$

$$v - v_0 = 4t + \frac{3}{2}t^2$$

$$v(t) = v_0 + 4t + 1.5t^2 = 5 + 4t + 1.5t^2 \quad (\mathrm{m \cdot s^{-1}})$$

At $t = 10\ \mathrm{s}$, the velocity is:

$$v(10) = 5 + 4 \times 10 + 1.5 \times 10^2 = 195\ \mathrm{m \cdot s^{-1}}$$

From $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ we get:

$$\mathrm{d}x = v\,\mathrm{d}t = (5 + 4t + 1.5t^2)\,\mathrm{d}t$$

Integrating both sides:

$$\int_{x_0}^{x} \mathrm{d}x = \int_{0}^{t} (5 + 4t + 1.5t^2)\,\mathrm{d}t$$

$$x - x_0 = 5t + 2t^2 + 0.5t^3$$

$$x(t) = x_0 + 5t + 2t^2 + 0.5t^3 = 2 + 5t + 2t^2 + 0.5t^3 \quad (\mathrm{m})$$

At $t = 10\ \mathrm{s}$, the position is:

$$x(10) = 2 + 5 \times 10 + 2 \times 10^2 + 0.5 \times 10^3 = 752\ \mathrm{m}$$

Solution walkthrough

1. **Core idea (applying calculus to kinematics)**:
    - **Differentiation direction**: position x —(differentiate)→ velocity v —(differentiate)→ acceleration a.
    - **Integration direction**: acceleration a —(integrate)→ velocity v —(integrate)→ position x.
    - In this problem, the acceleration a is given as a function of time t, and the velocity and position are required, so we need to perform two consecutive integrations along the **integration direction**.
2. **Step 1: find the velocity** $v(t)$

- The definition of acceleration is $a = \frac{\mathrm{d}v}{\mathrm{d}t}$. Rearranging gives: $\mathrm{d}v = a\,\mathrm{d}t$ (the change of velocity over an extremely short time interval).
- Accumulate both sides (i.e. integrate): on the left, the upper limit is the velocity v at an arbitrary time and the lower limit is the initial velocity $v_0 = 5$; on the right, the time runs from 0 to an arbitrary time t.
- Carrying out the integral: $\int (4 + 3t)\mathrm{d}t = 4t + \frac{3}{2}t^2$. Substituting the limits and rearranging gives the velocity formula: $v(t) = 5 + 4t + 1.5t^2$.
- Substituting $t = 10\ \mathrm{s}$ directly gives the velocity $v(10) = 195\ \mathrm{m/s}$.

3. **Step 2: find the position $x(t)$**
   - The definition of velocity is $v = \frac{\mathrm{d}x}{\mathrm{d}t}$. In the same way, rearranging gives: $\mathrm{d}x = v\,\mathrm{d}t$.
   - Integrating both sides: on the left, from the initial position $x_0 = 2$ to x; on the right, from 0 to t.
   - Carrying out the integral: $\int (5 + 4t + 1.5t^2)\mathrm{d}t = 5t + 2t^2 + 0.5t^3$.
   - Rearranging gives the position formula: $x(t) = 2 + 5t + 2t^2 + 0.5t^3$.
   - Substituting $t = 10\ \mathrm{s}$ directly gives the position $x(10) = 752\ \mathrm{m}$.

Exercise 1-6

Answer

Given the acceleration $a(t) = -A\omega^2 \cos \omega t$. At $t = 0$, $v_0 = 0$ and $x_0 = A$.

From $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, we get:

$$\mathrm{d}v = -A\omega^2 \cos \omega t\,\mathrm{d}t$$

Integrating both sides:

$$\int_0^v \mathrm{d}v = \int_0^t -A\omega^2 \cos \omega t\,\mathrm{d}t$$

$$v = [-A\omega \sin \omega t]_0^t = -A\omega \sin \omega t$$

From $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, we get:

$$\mathrm{d}x = -A\omega \sin \omega t\,\mathrm{d}t$$

Integrating both sides:

$$\int_A^x \mathrm{d}x = \int_0^t -A\omega \sin \omega t\,\mathrm{d}t$$

$$x - A = [A \cos \omega t]_0^t = A \cos \omega t - A$$

$$x(t) = A \cos \omega t \quad (\mathrm{m})$$

Hence the kinematic equation of this particle is:

$$x = A \cos \omega t$$

Solution walkthrough

1. **What is a "kinematic equation"?**
    - A kinematic equation is a functional relation expressing the position coordinate as a function of time, i.e. $x = f(t)$.
2. **First integration (finding the velocity v)**:
    - The integral is $v = \int a \, \mathrm{d}t = \int -A\omega^2 \cos \omega t \, \mathrm{d}t$.
    - Hint: by the fundamental formula of calculus, the integral of a cosine is a sine, i.e. $\int \cos(kx) \mathrm{d}x = \frac{1}{k} \sin(kx)$.
    - So, integrating $-A\omega^2 \cos \omega t$ gives $-A\omega^2 \cdot \frac{1}{\omega} \sin \omega t = -A\omega \sin \omega t$.
    - Since $v = 0$ at $t = 0$ and $\sin(0) = 0$, the constant term is 0. This gives the velocity function $v(t) = -A\omega \sin \omega t$.
3. **Second integration (finding the position x)**:
    - Integrate the velocity once more: $x = \int v \, \mathrm{d}t = \int -A\omega \sin \omega t \, \mathrm{d}t$.
    - Hint: the integral of a sine is minus a cosine, i.e. $\int \sin(kx) \mathrm{d}x = -\frac{1}{k} \cos(kx)$.
    - So, integrating $-A\omega \sin \omega t$ gives $-A\omega \cdot \left(-\frac{1}{\omega} \cos \omega t\right) = A \cos \omega t$.
    - Pay attention to **substituting the limits of the definite integral**: Left side: $\int_A^x \mathrm{d}x = x - A$; Right side: $[A \cos \omega t]_0^t = A \cos \omega t - A \cos(0) = A \cos \omega t - A$.
    - Simplifying both sides, the $-A$ cancels, and the final result is: $x = A \cos \omega t$. This is the kinematic equation of the particle.

Chapter 2 Dynamics of a Particle

Exercise 2-7

Answer

Given $m = 1 \ \mathrm{kg}$, the force $F(t) = 3 + 4t$, and the initial velocity $v_0 = 10 \ \mathrm{m \cdot s^{-1}}$.

(1) Find the impulse I of the force during $0 \sim 2 \ \mathrm{s}$: By the definition of impulse:

$$I = \int_0^2 F \, \mathrm{d}t = \int_0^2 (3 + 4t) \, \mathrm{d}t$$

$$I = \left[3t + 2t^2\right]_0^2 = (3 \times 2 + 2 \times 2^2) - 0 = 14 \ \mathrm{N \cdot s}$$

(2) Find the velocity of the particle at $t = 2 \ \mathrm{s}$: By the impulse–momentum theorem:

$$I = \Delta p = mv - mv_0$$

$$14 = 1 \times v - 1 \times 10$$

$$v = 24 \ \mathrm{m \cdot s^{-1}}$$

Solution walkthrough

1. **What is impulse?**
    - In physics, the cumulative effect of a force over time is called **impulse**. For a constant force, the impulse is $I = F \cdot \Delta t$; for a force that varies with time, it must be computed with a definite integral: $I = \int F \, \mathrm{d}t$.
2. **Analysis of part (1)**:
    - The problem asks for the impulse during $0 \sim 2\ \mathrm{s}$, so we integrate the varying force $F = 3 + 4t$ over the time interval $[0, 2]$.
    - Carrying out the integral: $\int (3 + 4t)\mathrm{d}t = 3t + 2t^2$.
    - Substituting the upper limit $t = 2$ and the lower limit $t = 0$: $(3 \times 2 + 2 \times 4) - 0 = 6 + 8 = 14\ \mathrm{N \cdot s}$ (the unit of impulse is the newton-second).
3. **Analysis of part (2)**:
    - The **impulse–momentum theorem** states: the impulse of the net external force on an object equals the change of its momentum, i.e. I = Δp = p_final − p_initial = mv_final − mv_initial.
    - Since we found the impulse $I = 14\ \mathrm{N \cdot s}$ in part (1), we can write the equation directly: $14 = m(v - v_0)$.
    - Substituting the known values $m = 1\ \mathrm{kg}$ and $v_0 = 10\ \mathrm{m/s}$ and solving gives $14 = 1 \times (v - 10) \implies v = 24\ \mathrm{m/s}$.

Exercise 2-10

Answer

Given the net force $F(t) = 400 - \frac{4}{3} \times 10^5 t$ and the speed at the muzzle $v = 300\ \mathrm{m \cdot s^{-1}}$.

(1) Find the time t the bullet takes to travel the full length of the barrel: From the problem, the net force is zero exactly when the bullet leaves the muzzle, i.e.:

$$F(t) = 0 \implies 400 - \frac{4}{3} \times 10^5 t = 0$$

$$t = \frac{400 \times 3}{4 \times 10^5} = 3 \times 10^{-3}\ \mathrm{s}$$

(2) Find the impulse I of the force on the bullet while it is in the barrel:

$$I = \int_0^t F \, \mathrm{d}t = \int_0^{3\times10^{-3}} \left(400 - \frac{4}{3} \times 10^5 t\right) \mathrm{d}t$$

$$I = \left[400t - \frac{2}{3} \times 10^5 t^2\right]_0^{3\times10^{-3}}$$

$$I = 400 \times (3 \times 10^{-3}) - \frac{2}{3} \times 10^5 \times (9 \times 10^{-6})$$

$$I = 1.2 - 0.6 = 0.6\ \mathrm{N \cdot s}$$

(3) Find the mass m of the bullet: By the impulse–momentum theorem, taking the bullet initially at rest ($v_0 = 0$):

$$I = mv - mv_0 \implies I = mv$$

$$0.6 = m \times 300$$

$$m = 2 \times 10^{-3}\ \text{kg} = 2\ \text{g}$$

Solution walkthrough

1. **Part (1): find the time t**
   - The key to the problem is that "**the net force is exactly zero where the bullet leaves the muzzle**".
   - This means that at the instant t when the bullet exits the muzzle, the force is $F = 0$. So we set up the equation $F(t) = 0$ directly and solve $t = 0.003\ \text{s}$ (equivalently written as $3 \times 10^{-3}\ \text{s}$, i.e. 3 milliseconds).
2. **Part (2): find the impulse I**
   - We again use the integral definition of the impulse of a varying force; the interval of integration is the time the bullet spends moving inside the barrel, i.e. from 0 to $3 \times 10^{-3}\ \text{s}$.
   - The integrand $400 - \frac{4}{3} \times 10^5 t$ integrates to $400t - \frac{2}{3} \times 10^5 t^2$.
   - Substituting the upper limit 3×10^{-3}:
     - The first term: $400 \times 0.003 = 1.2$;
     - The second term: $\frac{2}{3} \times 10^5 \times 9 \times 10^{-6} = \frac{2}{3} \times 0.9 = 0.6$;
     - Subtracting gives $1.2 - 0.6 = 0.6\ \text{N} \cdot \text{s}$.
3. **Part (3): find the mass m of the bullet**
   - Inside the barrel, the bullet accelerates from rest ($v_0 = 0$) to the muzzle speed $v = 300\ \text{m/s}$.
   - Applying the impulse–momentum theorem: the impulse $I = \Delta p = m \cdot v$.
   - With $I = 0.6\ \text{N} \cdot \text{s}$ and $v = 300\ \text{m/s}$, we get $m = \frac{I}{v} = \frac{0.6}{300} = 0.002\ \text{kg} = 2\ \text{g}$.

Exercise 2-16

Answer

Given $m = 2\ \text{kg}$ and the varying force $F_x(x) = 4 + 6x$. At $t = 0$, $x_0 = 0$ and $v_0 = 0$.

(1) Find the work W done by the force on the object during the motion from $x = 0$ to $x = 4\ \text{m}$: By the definition of work:

$$W = \int_{x_0}^{x_1} F_x\,\text{dx} = \int_0^4 (4 + 6x)\,\text{dx}$$

$$W = \left[4x + 3x^2\right]_0^4 = (4 \times 4 + 3 \times 4^2) - 0 = 16 + 48 = 64\ \text{J}$$

(2) Find the velocity v of the object at $x = 4\ \text{m}$: By the work–energy theorem:

$$W = \Delta E_k = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

$$64 = \frac{1}{2} \times 2 \times v^2 - 0 \implies v^2 = 64$$

$$v = 8\ \mathrm{m}\cdot\mathrm{s}^{-1}$$

(3) Find the impulse I of the varying force during the motion from $x = 0$ to $x = 4\ \mathrm{m}$: By the impulse–momentum theorem:

$$I = \Delta p = mv - mv_0$$

$$I = 2 \times 8 - 0 = 16\ \mathrm{N}\cdot\mathrm{s}$$

Solution walkthrough

1. **Part (1): how to compute the work of a varying force**
   - The **definition of work** is force times displacement. When the force F_x varies with the position x, the work must be computed by integrating over position, with the formula $W = \int F_x\,\mathrm{d}x$.
   - In this problem, the interval of integration is the position x going from 0 to $4\ \mathrm{m}$.
   - Carrying out the integral: $\int (4 + 6x)\mathrm{d}x = 4x + 3x^2$. Substituting the upper limit 4 gives the work $W = 64\ \mathrm{J}$ (joules).
2. **Part (2): find the velocity v**
   - The **work–energy theorem** states: the work done by the net external force on an object equals the change of its kinetic energy (ΔE_k = E_k,final − E_k,initial).
   - Since the object is initially at rest ($v_0 = 0$), the initial kinetic energy is 0.
   - Writing the equation: $W = \frac{1}{2}mv^2 \implies 64 = \frac{1}{2} \times 2 \times v^2 \implies v^2 = 64$, which easily gives the velocity at $4\ \mathrm{m}$ as $8\ \mathrm{m/s}$.
3. **Part (3): find the impulse I**
   - Computing this problem from the impulse definition $I = \int F\mathrm{d}t$ would be very complicated, because we do not know the force as a function of time t.
   - **Shortcut**: use the **impulse–momentum theorem** directly — the impulse equals the change of momentum $\Delta p = mv - mv_0$.
   - Here $m = 2\ \mathrm{kg}$, the final velocity is the $8\ \mathrm{m/s}$ just found, and the initial velocity is 0.
   - We immediately get: $I = 2 \times 8 = 16\ \mathrm{N}\cdot\mathrm{s}$. Using the relation between kinetic energy and momentum avoids the complicated integral.

Exercise 2-18

Answer

Given $m = 0.5\ \mathrm{kg}$ and the equations of motion $x = 5t$, $y = 0.5t^2$.

Differentiating with respect to time t gives the velocity components:

$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t} = 5\ \mathrm{m}\cdot\mathrm{s}^{-1}$$

$$v_y = \frac{\mathrm{d}y}{\mathrm{d}t} = t\ \mathrm{m}\cdot\mathrm{s}^{-1}$$

At $t_1 = 2\ \mathrm{s}$, the velocity components and the squared speed are:

$$v_{x1} = 5\ \mathrm{m}\cdot\mathrm{s}^{-1}, \quad v_{y1} = 2\ \mathrm{m}\cdot\mathrm{s}^{-1}$$

$$v_1^2 = v_{x1}^2 + v_{y1}^2 = 5^2 + 2^2 = 29\ \mathrm{m}^2\cdot\mathrm{s}^{-2}$$

At $t_2 = 4\ \mathrm{s}$, the velocity components and the squared speed are:

$$v_{x2} = 5\ \mathrm{m}\cdot\mathrm{s}^{-1}, \quad v_{y2} = 4\ \mathrm{m}\cdot\mathrm{s}^{-1}$$

$$v_2^2 = v_{x2}^2 + v_{y2}^2 = 5^2 + 4^2 = 41\ \mathrm{m}^2\cdot\mathrm{s}^{-2}$$

By the work–energy theorem, the work W done by the external force is:

$$W = \Delta E_k = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$W = \frac{1}{2} \times 0.5 \times (41 - 29) = 0.25 \times 12 = 3\ \mathrm{J}$$

Solution walkthrough

1. **Core method (a clever use of the work–energy theorem)**:
    - The problem asks for the work done by the external force during $2\ \mathrm{s} \sim 4\ \mathrm{s}$. The work done by the external force directly equals the **change of the kinetic energy** (work–energy theorem). So we only need the speed of the particle at the two instants $t = 2\ \mathrm{s}$ and $t = 4\ \mathrm{s}$.
2. **Step 1: find the velocity components**
    - The particle moves in two dimensions in the plane, and its velocity is composed of the x direction and the y direction together.
    - The x-velocity v_x is the derivative of the position $x = 5t$ with respect to time t; the result is the constant value 5.
    - The y-velocity v_y is the derivative of the position $y = 0.5t^2$ with respect to time t; the result is t.
3. **Step 2: compute the squared speeds at the two instants**
    - Formula for the squared speed (Pythagorean theorem): $v^2 = v_x^2 + v_y^2$.
    - At $t = 2\ \mathrm{s}$: $v_1^2 = 5^2 + 2^2 = 29$.
    - At $t = 4\ \mathrm{s}$: $v_2^2 = 5^2 + 4^2 = 41$.
4. **Step 3: find the work W**
    - Change of kinetic energy: $W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}m(v_2^2 - v_1^2)$.
    - Substituting the data: $W = \frac{1}{2} \times 0.5 \times (41 - 29) = 0.25 \times 12 = 3\ \mathrm{J}$.

Chapter 3 Rotation of a Rigid Body about a Fixed Axis

Exercise 3-6

> **Supplementary note:** In the original exam problem, the specific quantities to find after "Find:" are missing. Usually this kind of classic problem has two parts: (1) the **angular acceleration** β at the instant

> of release; (2) the **angular velocity** ω when the rod has rotated to the horizontal position. The standard solutions for these two classic quantities are given below.

Answer

Given $l = 1\ \mathrm{m}$, the moment of inertia about the end point O is $J = \frac{1}{3}ml^2$, and the initial release angle is $\theta = 60^\circ$.

```
   / (release position, 60 degrees above the horizontal)
  /
 /_ 60°
O ------------------- (horizontal position)
```

(1) Find the angular acceleration β at the instant of release: At the instant of release, the gravity mg acts at the center of mass of the rod (the midpoint $\frac{l}{2}$), and its moment arm about the axis O is $d = \frac{l}{2}\cos 60^\circ$. By the law of rotation $M = J\beta$:

$$mg\frac{l}{2}\cos 60^\circ = \frac{1}{3}ml^2\beta$$

$$\frac{1}{2}g\cos 60^\circ = \frac{1}{3}l\beta$$

$$\beta = \frac{3g\cos 60^\circ}{2l} = \frac{3 \times 9.8 \times 0.5}{2 \times 1} = 7.35\ \mathrm{rad \cdot s^{-2}}$$

(2) Find the angular velocity ω when the rod has rotated to the horizontal position: Take the horizontal plane containing the point O as the zero level of gravitational potential energy. By the law of conservation of mechanical energy:

$$E_{k1} + E_{p1} = E_{k2} + E_{p2}$$

$$0 + mg\frac{l}{2}\sin 60^\circ = \frac{1}{2}J\omega^2 + 0$$

$$mg\frac{l}{2}\sin 60^\circ = \frac{1}{2}\left(\frac{1}{3}ml^2\right)\omega^2$$

$$g\sin 60^\circ = \frac{1}{3}l\omega^2$$

$$\omega = \sqrt{\frac{3g\sin 60^\circ}{l}} = \sqrt{\frac{3 \times 9.8 \times \frac{\sqrt{3}}{2}}{1}} = \sqrt{25.46} \approx 5.05\ \mathrm{rad \cdot s^{-1}}$$

Solution walkthrough

1. **Part (1): applying the law of rotation of a rigid body (finding the angular acceleration)**
   - Just as particle motion uses $F = ma$, rigid-body rotation uses the **law of rotation**: $M = J\beta$, where M is the net external torque, J is the moment of inertia, and β is the angular acceleration of the rotation.
   - The gravity mg acts at the geometric center of the rod (i.e. the midpoint $l/2$). Gravity points vertically downward and the pivot O is at the left end, so the moment arm (the perpendicular distance from the axis to the line of action of the force) is $\frac{l}{2}\cos 60^\circ$.

- Setting up the equation: the gravity torque $M = mg \cdot \frac{l}{2}\cos 60^\circ$ equals $J\beta = \frac{1}{3}ml^2\beta$.
- Canceling the mass m and one length l on both sides and substituting the data, we get $\beta = 7.35\ \mathrm{rad/s^2}$.

2. **Part (2): conservation of mechanical energy (finding the angular velocity when the rod reaches the horizontal)**
   - As the rod swings down, only gravity does work, so **mechanical energy is conserved** (kinetic energy + potential energy = constant).
   - The rotational kinetic energy of a rigid body is given by $E_k = \frac{1}{2}J\omega^2$.
   - Change of the height of the center of mass: initially the center of mass is at the midpoint, tilted upward at 60°, so its height is $h = \frac{l}{2}\sin 60^\circ$; when the rod reaches the horizontal position, the height of the center of mass is 0.
   - The decrease of the gravitational potential energy equals the increase of the rotational kinetic energy: $mg\Delta h = \frac{1}{2}J\omega^2$.
   - Substituting the data gives the angular velocity $\omega \approx 5.05\ \mathrm{rad/s}$.

Exercise 3-8

Answer

Given $m_1 = 0.46\ \mathrm{kg}$, $m_2 = 0.50\ \mathrm{kg}$, $R = 0.05\ \mathrm{m}$; starting from rest, the weight descends the height $h = 0.75\ \mathrm{m}$ within $t = 5.0\ \mathrm{s}$.

The weight moves with uniformly accelerated rectilinear motion:

$$h = \frac{1}{2}at^2 \implies 0.75 = \frac{1}{2}a \times 5.0^2$$

$$a = \frac{2 \times 0.75}{25} = 0.06\ \mathrm{m \cdot s^{-2}}$$

The angular acceleration β of the pulley is:

$$\beta = \frac{a}{R} = \frac{0.06}{0.05} = 1.2\ \mathrm{rad \cdot s^{-2}}$$

Write the dynamic equations for m_2, m_1, and the pulley respectively (let the string tensions be T_2 and T_1 respectively, and take $g = 9.8\ \mathrm{m \cdot s^{-2}}$): For m_2 (descending):

$$m_2 g - T_2 = m_2 a \implies T_2 = m_2(g - a)$$

For m_1 (rising):

$$T_1 - m_1 g = m_1 a \implies T_1 = m_1(g + a)$$

For the pulley (rotating):

$$(T_2 - T_1)R = J\beta$$

Substituting T_2 and T_1 into the pulley equation:

$$\left[m_2(g - a) - m_1(g + a)\right] R = J\frac{a}{R}$$

$$J = \frac{R^2}{a}\left[(m_2 - m_1)g - (m_2 + m_1)a\right]$$

Substituting the numerical values:

$$J = \frac{0.05^2}{0.06}\left[(0.50 - 0.46) \times 9.8 - (0.50 + 0.46) \times 0.06\right]$$

$$J = \frac{0.0025}{0.06}\left[0.04 \times 9.8 - 0.96 \times 0.06\right]$$

$$J = \frac{1}{24}\left[0.392 - 0.0576\right] = \frac{1}{24} \times 0.3344 \approx 0.0139\ \mathrm{kg \cdot m^2}$$

Solution walkthrough

1. **Step 1: find the acceleration a and the angular acceleration β**
    - The falling weight undergoes uniformly accelerated rectilinear motion. Given the initial velocity 0 and that it descends $h = 0.75\ \mathrm{m}$ within the time $t = 5\ \mathrm{s}$.
    - Using the kinematic formula $h = \frac{1}{2}at^2$, we directly find the acceleration $a = 0.06\ \mathrm{m/s^2}$.
    - Since the string does not slip on the pulley, the linear acceleration of the pulley rim equals the translational acceleration a of the weight, so the angular acceleration of the pulley's rotation is $\beta = \frac{a}{R} = 1.2\ \mathrm{rad/s^2}$.
2. **Step 2: analyze the forces on the system (the core!)**
    - **Note**: because the pulley has mass and moment of inertia, the string tensions T_1 and T_2 on the two sides are **not equal** (the m_2 on the right is heavier, so $T_2 > T_1$, and the two tensions spin the pulley with increasing clockwise speed).
    - For the left m_2: it accelerates downward, so gravity is greater than the tension: $m_2 g - T_2 = m_2 a$.
    - For the right m_1: it accelerates upward, so the tension is greater than gravity: $T_1 - m_1 g = m_1 a$.
    - For the pulley: the torques come from the clockwise torque $T_2 R$ of T_2 and the counterclockwise torque $T_1 R$ of T_1. The net torque is $(T_2 - T_1)R$. By the law of rotation, $(T_2 - T_1)R = J\beta$.
3. **Step 3: solve the system by elimination to find J**
    - Express T_2 and T_1 from the first two equations respectively, and substitute them into the third equation.
    - Rearrange into an expression for the moment of inertia J, substitute the given mass, radius, gravitational acceleration $g = 9.8$, and the computed a, and the calculation gives $J \approx 0.0139\ \mathrm{kg \cdot m^2}$.

Exercise 3-10

Answer

Given $m_1 = 5\ \mathrm{kg}$, $\theta = 30^\circ$, $\mu = 0.25$, $m_2 = 10\ \mathrm{kg}$, pulley $m = 20\ \mathrm{kg}$, radius $R = 0.2\ \mathrm{m}$, and the pulley is a disk ($J = \frac{1}{2}mR^2$).

Let the magnitude of the acceleration of the weights be a (m_2 accelerates downward, m_1 accelerates up the incline), and the angular acceleration of the pulley be $\beta = \frac{a}{R}$. Let the string tension on the m_1 side be T_1 and

the string tension on the m_2 side be T_2.

Write the dynamic equations for each object:

1. For m_2 (accelerating vertically downward):

$$m_2 g - T_2 = m_2 a \quad \text{— (1)}$$

2. For m_1 (accelerating up the incline):

$$T_1 - m_1 g \sin\theta - f = m_1 a \quad \text{— (2)}$$

   where the friction force is $f = \mu N = \mu m_1 g \cos\theta$.

3. For the disk pulley (rotating clockwise):

$$(T_2 - T_1) R = J\beta = \left(\frac{1}{2} m R^2\right)\left(\frac{a}{R}\right) \implies T_2 - T_1 = \frac{1}{2} m a \quad \text{— (3)}$$

Adding equations (1) and (2), and using equation (3):

$$(m_2 g - T_2) + (T_1 - m_1 g \sin\theta - \mu m_1 g \cos\theta) = (m_1 + m_2) a$$

$$m_2 g - (T_2 - T_1) - m_1 g(\sin\theta + \mu\cos\theta) = (m_1 + m_2) a$$

$$m_2 g - \frac{1}{2} m a - m_1 g(\sin\theta + \mu\cos\theta) = (m_1 + m_2) a$$

Rearranging gives the formula for the acceleration a:

$$a = \frac{m_2 - m_1(\sin\theta + \mu\cos\theta)}{m_1 + m_2 + \frac{1}{2}m} g$$

Substituting the data (taking $g = 9.8\ \mathrm{m \cdot s^{-2}}$):

$$a = \frac{10 - 5 \times (\sin 30^\circ + 0.25 \cos 30^\circ)}{5 + 10 + \frac{1}{2} \times 20} \times 9.8$$

$$a = \frac{10 - 5 \times (0.5 + 0.25 \times 0.866)}{25} \times 9.8 = \frac{10 - 5 \times 0.7165}{25} \times 9.8$$

$$a = \frac{10 - 3.58}{25} \times 9.8 = \frac{6.42}{25} \times 9.8 \approx 2.52\ \mathrm{m \cdot s^{-2}}$$

Find the string tensions: Substituting a into equation (1) to find T_2:

$$T_2 = m_2(g - a) = 10 \times (9.8 - 2.52) = 72.8\ \mathrm{N}$$

Substituting a into equation (3) to find T_1:

$$T_1 = T_2 - \frac{1}{2} m a = 72.8 - \frac{1}{2} \times 20 \times 2.52 = 72.8 - 25.2 = 47.6\ \mathrm{N}$$

Solution walkthrough

1. **Analysis of the overall motion tendency**:
   - The hanging weight on the left, $m_2 = 10\ \mathrm{kg}$, is heavier, while the $m_1 = 5\ \mathrm{kg}$ on the incline is lighter and is held back by the incline. Therefore, the system must move to the left, i.e. m_2 accelerates downward and m_1 slides up the incline toward the upper right.
2. **Set up the dynamic equations for each part**:

- **For the m_1 on the incline**: along the incline direction, it feels the upward pull T_1 of the string, the component of gravity down the incline $m_1 g \sin\theta$, and the opposing friction force f that hinders the motion. By the friction formula: $f = \mu N$, where the normal force perpendicular to the incline is $N = m_1 g \cos\theta$. So $f = \mu m_1 g \cos\theta$. Newton's second law gives: $T_1 - m_1 g \sin\theta - \mu m_1 g \cos\theta = m_1 a$.
- **For the hanging m_2**: it feels the downward gravity $m_2 g$ and the upward pull T_2 of the string. Equation: $m_2 g - T_2 = m_2 a$.
- **For the disk pulley**: the pulls on the left and right are not equal, producing a driving torque. $(T_2 - T_1)R = J\beta$. Because the pulley is a uniform disk, its moment of inertia is $J = \frac{1}{2}mR^2$; substituting $\beta = \frac{a}{R}$ and canceling the radius R simplifies this to: $T_2 - T_1 = \frac{1}{2}ma$.

3. **Solve the simultaneous equations**:
   - This is a system of three equations with the unknowns a, T_1, and T_2.
   - The fastest way is to rearrange the first two equations and substitute them into the third one, which eliminates the tensions directly and gives the acceleration $a \approx 2.52\ \mathrm{m/s^2}$.
   - Once the acceleration is found, substitute it back into the step-by-step equations above, and the two string tensions come out easily: $T_1 = 47.6\ \mathrm{N}$ and $T_2 = 72.8\ \mathrm{N}$.

Chapter 6 Mechanical Oscillations and Mechanical Waves

Exercise 6-5

Answer

Given the period $T = 2\ \mathrm{s}$, the amplitude $A = 0.06\ \mathrm{m}$, and the wave speed $u = 2\ \mathrm{m \cdot s^{-1}}$. At $t = 0$, $y_0 = -A = -0.06\ \mathrm{m}$.

(1) Find the kinematic equation of this particle: The angular frequency ω is:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi\ \mathrm{rad \cdot s^{-1}}$$

Let the simple harmonic oscillation equation be:

$$y = A\cos(\omega t + \phi) = 0.06\cos(\pi t + \phi)$$

Substituting the initial condition $t = 0, y = -0.06\ \mathrm{m}$:

$$-0.06 = 0.06\cos\phi \implies \cos\phi = -1 \implies \phi = \pi$$

Hence the kinematic equation of this particle is:

$$y = 0.06\cos(\pi t + \pi) \quad (\mathrm{m})$$

(2) Find the wave expression of the one-dimensional simple harmonic wave propagating along the positive x-axis: The wave propagates along the positive x-axis, so the oscillation phase of every particle lags behind. The wave expression is:

$$y(x,t) = A\cos\left[\omega\left(t - \frac{x}{u}\right) + \phi\right]$$

Substituting $A = 0.06$, $\omega = \pi$, $u = 2$, $\phi = \pi$:

$$y(x,t) = 0.06\cos\left[\pi\left(t - \frac{x}{2}\right) + \pi\right] \quad (\mathrm{m})$$

(3) Find the wavelength λ of this wave:

$$\lambda = uT = 2 \times 2 = 4\ \mathrm{m}$$

Solution walkthrough

1. **The basic form of the simple harmonic motion equation**:
   - The general formula of simple harmonic motion is $y = A\cos(\omega t + \phi)$, where:
     - A is the amplitude (the maximum displacement, given as 0.06).
     - ω is the angular frequency, related to the period T by $\omega = \frac{2\pi}{T}$, giving $\omega = \pi$.
     - ϕ is the initial phase (the angular state at the instant $t = 0$).
2. **Part (1): find the initial phase ϕ**
   - The problem says "at $t = 0$ the particle is exactly at the maximum displacement in the negative direction", which means the displacement is $y = -A = -0.06\ \mathrm{m}$.
   - Substituting into the simple harmonic formula: $-0.06 = 0.06\cos\phi \implies \cos\phi = -1 \implies \phi = \pi$.
   - So the equation of motion of this particle is $y = 0.06\cos(\pi t + \pi)$ (using a trigonometric identity it can also be simplified to $y = -0.06\cos(\pi t)$).
3. **Part (2): find the wave expression of the simple harmonic wave**
   - A wave is the propagation of an oscillation through space. If a wave propagates in the **positive** direction of the x-axis, then the oscillation of the particle at the distance x from the origin **lags behind** the origin by the time $\Delta t = \frac{x}{u}$ (distance divided by wave speed).
   - So we only need to replace the time t in the oscillation equation of the particle at the origin with $\left(t - \frac{x}{u}\right)$ — that's all!
   - Direct replacement: $y = 0.06\cos\left[\pi\left(t - \frac{x}{2}\right) + \pi\right]$. This is the very classic way of writing the wave expression.
4. **Part (3): find the wavelength λ**
   - The wavelength is the distance the wave travels in one period: λ = wave speed × period = u × T.
   - Given $u = 2\ \mathrm{m/s}$ and $T = 2\ \mathrm{s}$, the wavelength is $\lambda = 4\ \mathrm{m}$.

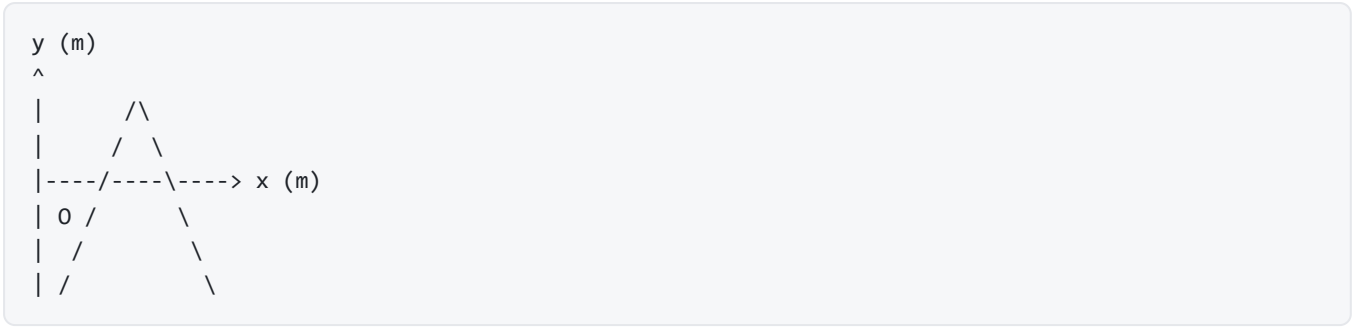
Exercise 6-10

Answer

From the waveform graph at $t = 0$ we can read off:

- Amplitude $A = 0.04\ \mathrm{m}$

- Wavelength $\lambda = 0.40\ \mathrm{m}$
- Wave speed $u = 0.08\ \mathrm{m \cdot s^{-1}}$ (propagating in the positive x-axis direction)



The period T and the angular frequency ω are:

$$T = \frac{\lambda}{u} = \frac{0.40}{0.08} = 5.0\ \mathrm{s}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{5} = 0.4\pi\ \mathrm{rad \cdot s^{-1}}$$

(1) Find the wave expression of this wave: Let the oscillation equation of the particle at the point O ($x = 0$) be $y_O = A\cos(\omega t + \phi_0)$. From the waveform graph at $t = 0$, the displacement of the particle at the origin O is $y_O(0) = 0$. Since the wave propagates in the positive x direction and the waveform moves to the right, the particle at O is moving in the negative y-axis direction at this moment, i.e. the oscillation velocity satisfies $v_y < 0$. By the method for determining the initial phase of simple harmonic motion:

$$\begin{cases} \cos\phi_0 = 0 \\ \sin\phi_0 > 0 \end{cases} \Longrightarrow\ \phi_0 = \frac{\pi}{2}$$

So the oscillation equation of the origin O is:

$$y_O = 0.04\cos\left(0.4\pi t + \frac{\pi}{2}\right)\quad (\mathrm{m})$$

Since the wave propagates along the positive x-axis, the wave expression is:

$$y(x,t) = A\cos\left[\omega\left(t - \frac{x}{u}\right) + \phi_0\right]$$

$$y(x,t) = 0.04\cos\left[0.4\pi\left(t - \frac{x}{0.08}\right) + \frac{\pi}{2}\right]$$

$$y(x,t) = 0.04\cos\left(0.4\pi t - 5\pi x + \frac{\pi}{2}\right)\quad (\mathrm{m})$$

(2) Find the equation of motion of the particle at the point P ($x = 0.20\ \mathrm{m}$): Substituting $x = 0.20\ \mathrm{m}$ into the wave expression:

$$y_P(t) = 0.04\cos\left(0.4\pi t - 5\pi \times 0.20 + \frac{\pi}{2}\right)$$

$$y_P(t) = 0.04\cos\left(0.4\pi t - \pi + \frac{\pi}{2}\right)$$

$$y_P(t) = 0.04\cos\left(0.4\pi t - \frac{\pi}{2}\right)\quad (\mathrm{m})$$

Solution walkthrough

1. **Step 1: extract the useful information from the waveform graph (t=0)**

- **Amplitude** A: the maximum height of a wave crest, which is $0.04\ \mathrm{m}$ on the y-axis.
- **Wavelength** λ: the length of one complete waveform; in the graph, one complete cycle of the wave spans $0.40\ \mathrm{m}$ on the x-axis.
- **Wave speed** u: the graph directly gives $u = 0.08\ \mathrm{m/s}$.
- Hence the period is $T = \frac{\lambda}{u} = 5\ \mathrm{s}$ and the angular frequency is $\omega = \frac{2\pi}{T} = 0.4\pi$.

2. **Step 2: find the initial phase ϕ_0 of the origin O**
   - Look at the position of the origin O at $t = 0$: it is exactly at $y = 0$ (the equilibrium position). This means $\cos\phi_0 = 0$, so the initial phase could be $\frac{\pi}{2}$ or $-\frac{\pi}{2}$.
   - **Judging the direction of motion by shifting the waveform**: since the wave travels to the right, shift the waveform slightly to the right, and you will find that the particle at the origin O is about to **swing downward** (its velocity points in the negative direction).
   - The velocity points in the negative direction, which means the initial phase ϕ_0 lies in the first or second quadrant; hence $\phi_0 = \frac{\pi}{2}$.
3. **Step 3: write out the wave expression**
   - Apply the formula for a right-traveling wave: $y = A\cos\left[\omega\left(t - \frac{x}{u}\right) + \phi_0\right]$.
   - Substituting the data and simplifying: $0.4\pi\left(t - \frac{x}{0.08}\right) + \frac{\pi}{2} = 0.4\pi t - 5\pi x + \frac{\pi}{2}$.
4. **Step 4: find the equation of motion at the specific point P**
   - The wave expression gives the displacement of **any point** in space at **any time**.
   - Since the equation at the point P (coordinate $x = 0.20\ \mathrm{m}$) is required, we only need to substitute $x = 0.20$ into the wave expression and simplify the constant terms, which gives a one-dimensional oscillation equation containing only the time t: $y_P = 0.04\cos(0.4\pi t - \frac{\pi}{2})$.

Chapter 9 Electrostatic Field

Exercise 9-9

Answer

Given two imaginary concentric spherical surfaces of radii $R_1 = 0.1\ \mathrm{m}$ and $R_2 = 0.2\ \mathrm{m}$ respectively, with volume charge density $\rho = 10^{-9}\ \mathrm{C\cdot m^{-3}}$.

By the spherical symmetry, choose a concentric spherical surface as the Gaussian surface. Gauss's law takes the form:

$$\oint_S E\,\mathrm{d}S = E\cdot 4\pi r^2 = \frac{\sum q_{\text{enclosed}}}{\varepsilon_0}$$

$$E = \frac{\sum q_{\text{enclosed}}}{4\pi\varepsilon_0 r^2}$$

(1) When $r = 5\ \mathrm{cm} = 0.05\ \mathrm{m}$: Since $r < R_1$, the charge enclosed by the Gaussian surface is Σq_enc = 0. Hence:

$$E_1 = 0$$

(2) When $r = 15\ \mathrm{cm} = 0.15\ \mathrm{m}$: Since $R_1 < r < R_2$, the charge enclosed by the Gaussian surface is the charge of the spherical shell with radii from R_1 to r:

$$\sum q_{\text{enclosed}} = \rho V' = \rho \cdot \frac{4}{3}\pi(r^3 - R_1^3)$$

Hence the electric field strength there is:

$$E_2 = \frac{\rho \cdot \frac{4}{3}\pi(r^3 - R_1^3)}{4\pi\varepsilon_0 r^2} = \frac{\rho(r^3 - R_1^3)}{3\varepsilon_0 r^2}$$

Substituting the data (taking $\varepsilon_0 \approx 8.85 \times 10^{-12}\ \mathrm{C^2 \cdot N^{-1} \cdot m^{-2}}$):

$$E_2 = \frac{10^{-9} \times (0.15^3 - 0.1^3)}{3 \times 8.85 \times 10^{-12} \times 0.15^2} = \frac{10^{-9} \times (0.003375 - 0.001)}{3 \times 8.85 \times 10^{-12} \times 0.0225}$$

$$E_2 = \frac{10^{-9} \times 0.002375}{5.97375 \times 10^{-13}} \approx 3.98\ \mathrm{V \cdot m^{-1}}$$

(3) When $r = 50\ \mathrm{cm} = 0.5\ \mathrm{m}$: Since $r > R_2$, the Gaussian surface encloses the charge of the entire charged spherical shell:

$$\sum q_{\text{enclosed}} = \rho \cdot \frac{4}{3}\pi(R_2^3 - R_1^3)$$

Hence the electric field strength there is:

$$E_3 = \frac{\rho(R_2^3 - R_1^3)}{3\varepsilon_0 r^2}$$

Substituting the data:

$$E_3 = \frac{10^{-9} \times (0.2^3 - 0.1^3)}{3 \times 8.85 \times 10^{-12} \times 0.5^2} = \frac{10^{-9} \times (0.008 - 0.001)}{3 \times 8.85 \times 10^{-12} \times 0.25}$$

$$E_3 = \frac{7 \times 10^{-12}}{6.6375 \times 10^{-12}} \approx 1.05\ \mathrm{V \cdot m^{-1}}$$

Solution walkthrough

1. **The core tool: Gauss's law**
    - In an electrostatic field, if the charge distribution is highly symmetric (e.g. a sphere, a spherical shell, an infinitely long cylinder), the one and only way to find the electric field strength is **Gauss's law**.
    - Formula of Gauss's law: the electric flux $\oint E\mathrm{d}S$ through a closed surface (the Gaussian surface) equals the total net charge Σq_enc enclosed inside that surface divided by the vacuum permittivity ε_0.
    - For a spherically symmetric distribution, we choose a concentric spherical surface of radius r as the Gaussian surface. On this surface, the magnitude of the field strength E is the same everywhere and perpendicular to the surface, so the integral part simplifies directly to $E \cdot (4\pi r^2)$ (the field times the area of the sphere).
    - The general solving formula is: E = Σq_enc / (4πε₀r²). We only need to analyze **how much charge is enclosed by the Gaussian surfaces of different radii**.
2. **Part (1): $r = 5\ \mathrm{cm}$ (inside the inner sphere)**

- All the charge is distributed in the layer between $10\ \mathrm{cm} \sim 20\ \mathrm{cm}$.
- The Gaussian surface of radius $5\ \mathrm{cm}$ encloses nothing at all, so the enclosed charge is q_enc = 0. The electric field strength inside is therefore simply 0.

3. **Part (2):** $r = 15\ \mathrm{cm}$ **(in the middle)**
   - This Gaussian surface encloses part of the charge. Specifically, it encloses the thickness of the charged spherical shell from the inner radius $10\ \mathrm{cm}$ ($0.1\ \mathrm{m}$) up to the current radius $15\ \mathrm{cm}$ ($0.15\ \mathrm{m}$).
   - The charged volume: $V' = \frac{4}{3}\pi r^3 - \frac{4}{3}\pi R_1^3$. The charge is the volume times the volume charge density ρ.
   - Substituting into the Gauss's law formula and canceling the constants: $E = \frac{\rho(r^3 - R_1^3)}{3\varepsilon_0 r^2}$. Substituting the numbers gives $E \approx 3.98\ \mathrm{V/m}$.
4. **Part (3):** $r = 50\ \mathrm{cm}$ **(on the outside)**
   - The Gaussian surface is now very large and encloses the entire charged layer ($10\ \mathrm{cm} \sim 20\ \mathrm{cm}$).
   - The total enclosed charge is fixed and corresponds to the volume of the entire spherical shell: V_total = (4/3)π(R_2^3 − R_1^3).
   - Substituting into the Gauss's law formula in the same way: $E = \frac{\rho(R_2^3 - R_1^3)}{3\varepsilon_0 r^2}$. Substituting $r = 0.5\ \mathrm{m}$ and computing gives $E \approx 1.05\ \mathrm{V/m}$.

Exercise 9-12

Answer

By the axial symmetry of the "infinitely long" charged cylinder, choose a cylindrical surface of radius r and length L, coaxial with the cylinder, as the closed Gaussian surface.

Gauss's law:

$$\oint_S \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = E \cdot 2\pi r L = \frac{\sum q_{\text{enclosed}}}{\varepsilon_0}$$

(1) When $r < R$ (inside the cylinder): The charge enclosed by the Gaussian surface is:

$$\sum q_{\text{enclosed}} = \rho \cdot \pi r^2 L$$

Substituting into Gauss's law:

$$E \cdot 2\pi r L = \frac{\rho \cdot \pi r^2 L}{\varepsilon_0} \implies E = \frac{\rho}{2\varepsilon_0} r$$

(2) When $r \geq R$ (outside the cylinder): The Gaussian surface encloses all the charge of this segment of the cylinder:

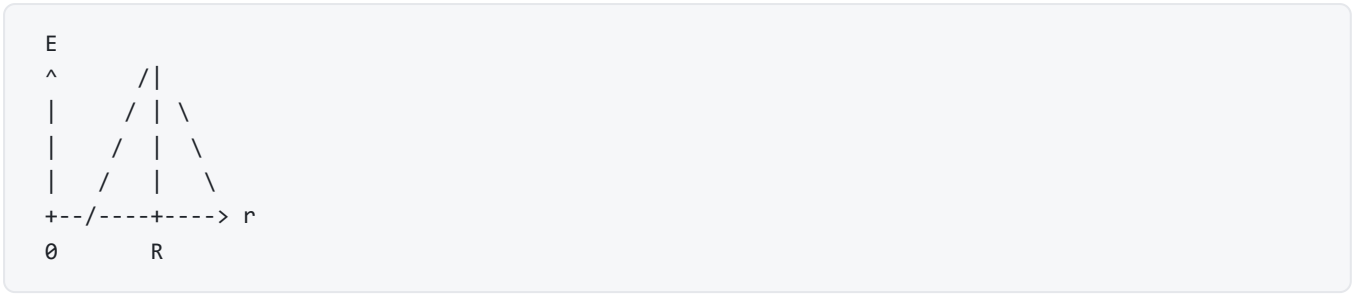
$$\sum q_{\text{enclosed}} = \rho \cdot \pi R^2 L$$

Substituting into Gauss's law:

$$E \cdot 2\pi r L = \frac{\rho \cdot \pi R^2 L}{\varepsilon_0} \implies E = \frac{\rho R^2}{2\varepsilon_0} \frac{1}{r}$$

(3) The E-r curve:

- In the range $r < R$, E is proportional to r (a linear rise).
- In the range $r \geq R$, E is inversely proportional to r (a hyperbolic fall).
- The field reaches its maximum at $r = R$: $E_{\max} = \frac{\rho R}{2\varepsilon_0}$.



Solution walkthrough

1. **How to construct the Gaussian surface? (Cylindrical symmetry)**
    - For the "infinitely long" cylinder, its field lines radiate horizontally outward, like a hedgehog's spines.
    - We build a coaxial **cylindrical tin can** of radius r and height L as the Gaussian surface.
    - The field passes only through the curved side of this "can", not through the top and bottom faces.
    - The curved side area of the can is $2\pi r L$, so the total electric flux through the Gaussian surface is $E \cdot (2\pi r L)$.
2. **Stage 1: inside the cylinder ($r < R$)**
    - The Gaussian tin can contains only part of the charged cylinder inside it.
    - The volume of this part of the cylinder is $\pi r^2 L$, so the charge inside it is q_enc = ρ · πr²L.
    - Substituting into Gauss's formula: $E \cdot 2\pi r L = \frac{\rho \cdot \pi r^2 L}{\varepsilon_0}$.
    - Canceling π, r, L on both sides gives $E = \frac{\rho}{2\varepsilon_0} r$.
    - **Physical meaning**: inside the cylinder, the field strength is proportional to the radius r; the farther out, the stronger the field (because more charge is enclosed).
3. **Stage 2: outside the cylinder ($r \geq R$)**
    - The Gaussian can is now thicker than the charged cylinder, so it contains the entire charged cylinder segment of radius R.
    - The charge it contains is fixed: q_enc = ρ · πR²L.
    - Substituting into Gauss's formula: $E \cdot 2\pi r L = \frac{\rho \cdot \pi R^2 L}{\varepsilon_0}$.
    - Canceling π, L on both sides gives $E = \frac{\rho R^2}{2\varepsilon_0} \frac{1}{r}$.
    - **Physical meaning**: outside the cylinder, the field strength is inversely proportional to the distance r; the farther away, the weaker the field.
4. **How to draw the E-r curve**:
    - From 0 to R, draw a straight line rising from the origin (a proportional function);

- Beyond R, draw a smooth descending curve that gradually approaches the horizontal axis (an inverse proportional function).

Exercise 9-22

Answer

Given two concentric spherical surfaces of radii $R_1 = 0.1\ \mathrm{m}$ and $R_2 = 0.3\ \mathrm{m}$ respectively. The small sphere carries $q_1 = 10^{-8}\ \mathrm{C}$ and the large sphere carries $q_2 = 1.5 \times 10^{-8}\ \mathrm{C}$.

By the spherical symmetry, use Gauss's law to find the field strength distribution $E(r)$ in each region:

1. When $r < R_1$:

$$E = 0$$

2. When $R_1 < r < R_2$:

$$E = \frac{q_1}{4\pi\varepsilon_0 r^2}$$

3. When $r > R_2$:

$$E = \frac{q_1 + q_2}{4\pi\varepsilon_0 r^2}$$

By the integral relation between the potential and the field strength, $V(r) = \int_r^\infty E\,\mathrm{d}r$ (taking the potential at infinity to be zero):

(1) Find the potential V_1 at $r = 20\ \mathrm{cm} = 0.2\ \mathrm{m}$ from the center: Since $R_1 < r < R_2$, the potential is a piecewise accumulated integral:

$$V_1 = \int_r^{R_2} E(R_1 < r < R_2)\,\mathrm{d}r + \int_{R_2}^\infty E(r > R_2)\,\mathrm{d}r$$

$$V_1 = \int_r^{R_2} \frac{q_1}{4\pi\varepsilon_0 r^2}\,\mathrm{d}r + \int_{R_2}^\infty \frac{q_1 + q_2}{4\pi\varepsilon_0 r^2}\,\mathrm{d}r$$

$$V_1 = \frac{q_1}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{R_2}\right) + \frac{q_1 + q_2}{4\pi\varepsilon_0 R_2}$$

$$V_1 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{R_2}\right)$$

Substituting the data (taking $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \mathrm{N \cdot m^2 \cdot C^{-2}}$):

$$V_1 = 9 \times 10^9 \times \left(\frac{10^{-8}}{0.2} + \frac{1.5 \times 10^{-8}}{0.3}\right)$$

$$V_1 = 9 \times 10^9 \times \left(5 \times 10^{-8} + 5 \times 10^{-8}\right) = 9 \times 10^9 \times 10^{-7} = 900\ \mathrm{V}$$

(2) Find the potential V_2 at $r = 50\ \mathrm{cm} = 0.5\ \mathrm{m}$ from the center: Since $r > R_2$, the field is determined only by the outside region:

$$V_2 = \int_r^\infty \frac{q_1 + q_2}{4\pi\varepsilon_0 r^2}\,\mathrm{d}r = \frac{q_1 + q_2}{4\pi\varepsilon_0 r}$$

Substituting the data:

$$V_2 = 9 \times 10^9 \times \frac{10^{-8} + 1.5 \times 10^{-8}}{0.5}$$

$$V_2 = 9 \times 10^9 \times \frac{2.5 \times 10^{-8}}{0.5} = 9 \times 10^9 \times 5 \times 10^{-8} = 450 \text{ V}$$

Solution walkthrough

1. **The nature of the potential (superposition principle and integral formula)**
    - The potential is a scalar and enjoys an extremely handy **superposition principle**.
    - For a uniformly charged spherical surface there is a very famous result:
        - If you are **outside** the spherical surface ($r > R$), the potential it produces at your location is identical to that of a point charge with all the charge concentrated at the center: $V = \frac{q}{4\pi\varepsilon_0 r}$.
        - If you are **inside** the spherical surface ($r < R$), the field strength at any point inside is zero, so the potential is the same at every interior point and equals the potential of the surface itself: $V = \frac{q}{4\pi\varepsilon_0 R}$.
    - Once you master this result, this problem needs **no complicated integral at all** — just superpose the formulas!
2. **Part (1): find the potential at $r = 20\ \text{cm}$**
    - This point is $20\ \text{cm}$ from the center, in the **middle** between the two spherical surfaces.
    - It lies **outside** the small sphere (radius $10\ \text{cm}$), so the potential produced here by the small sphere is computed with the outside-point formula: V_small = (1/4πε₀) · q_1/r.
    - It lies **inside** the large sphere (radius $30\ \text{cm}$), so the potential produced here by the large sphere equals the potential at its surface: V_large = (1/4πε₀) · q_2/R_2.
    - The total potential is the direct sum: V = V_small + V_large = 9 × 10^9 × (10^{-8}/0.2 + 1.5 × 10^{-8}/0.3) = 900 V.
3. **Part (2): find the potential at $r = 50\ \text{cm}$**
    - This point is $50\ \text{cm}$ from the center, **outside** both spherical surfaces.
    - Therefore it is an "outside point" with respect to both the small and the large sphere.
    - The total potential equals the sum of the outside-point potentials contributed by the two spheres: $V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1+q_2}{r}$.
    - Substituting and computing: $V = 9 \times 10^9 \times \frac{2.5\times10^{-8}}{0.5} = 450\ \text{V}$.

Chapter 10 Steady Magnetic Field

Exercise 10-9

Answer

Given a hollow cylindrical conductor with inner and outer radii a and b, carrying a uniformly distributed current I.

The current density j distributed over the cross-section is:

$$j = \frac{I}{\pi(b^2 - a^2)}$$

By the axial symmetry, choose a coaxial circle of radius r as the Ampèrian loop L. By Ampère's circuital law:

$$\oint_L \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = B \cdot 2\pi r = \mu_0 \sum I_{\text{enclosed}}$$

$$B = \frac{\mu_0 \sum I_{\text{enclosed}}}{2\pi r}$$

(1) Prove the formula for the magnetic induction in the region $a < r < b$: The current enclosed inside the loop L is only the current in the annular region of the cross-section with radii from a to r:

$$\sum I_{\text{enclosed}} = j \cdot \pi(r^2 - a^2) = \frac{I}{\pi(b^2 - a^2)} \cdot \pi(r^2 - a^2) = I\frac{r^2 - a^2}{b^2 - a^2}$$

Substituting into Ampère's circuital law:

$$B = \frac{\mu_0}{2\pi r}\left(I\frac{r^2 - a^2}{b^2 - a^2}\right) = \frac{\mu_0 I}{2\pi(b^2 - a^2)}\frac{r^2 - a^2}{r}$$

Q.E.D.

(2) Check the formula in the limiting case $a = 0$: When $a = 0$, the hollow cylinder becomes a solid, uniformly current-carrying cylinder of radius b. Substituting into the formula:

$$B = \frac{\mu_0 I}{2\pi(b^2 - 0)}\frac{r^2 - 0}{r} = \frac{\mu_0 I r}{2\pi b^2}$$

This agrees exactly with the formula for the magnetic induction inside a solid wire with uniform current, so the formula passes the check.

(3) When $r \geq b$: The Ampèrian loop encloses the entire current of the conductor, i.e. ΣI_enc = I. Substituting into Ampère's circuital law gives:

$$B \cdot 2\pi r = \mu_0 I \implies B = \frac{\mu_0 I}{2\pi r}$$

Solution walkthrough

1. **The core tool: Ampère's circuital law**
    - When finding the magnetic field produced by a symmetric current distribution, we use **Ampère's circuital law**: ∮B dl = μ₀ ΣI_enc.
    - Choose a concentric circular loop of radius r as the path; the integral on the left simplifies directly to: the magnetic induction B times the circumference $2\pi r$.
    - The circuital law simplifies to: B = μ₀ΣI_enc / (2πr). The key question is likewise: **how much current passes through the interior of the loop**?
2. **How to compute the partial current I_enc (the proportion method)**
    - The current is uniformly distributed over the solid part of the conductor (the annular cross-section).
    - The solid cross-sectional area of the conductor is: $\pi b^2 - \pi a^2 = \pi(b^2 - a^2)$.

- The circle of radius r we draw ($a < r < b$) encloses the solid cross-sectional area: $\pi r^2 - \pi a^2 = \pi(r^2 - a^2)$.
- Since the current is uniformly distributed, by the ratio of areas the enclosed current is:

$$\sum I_{\text{enclosed}} = I \times \frac{\text{enclosed area}}{\text{total cross-sectional area}} = I\frac{r^2 - a^2}{b^2 - a^2}$$

- Substituting this current into B = μ₀ΣI_enc / (2πr) and rearranging gives the formula the problem asks to prove.

3. **What does the limiting case $a = 0$ mean?**
   - $a = 0$ means the inner radius is zero, i.e. there is no hollow core any more and the conductor becomes an ordinary solid wire.
   - Substituting $a = 0$ into the proved formula gives $B = \frac{\mu_0 I r}{2\pi b^2}$, which matches the classic result that the magnetic field inside a solid wire is proportional to the distance.
4. **When $r \geq b$ (outside the wire)**:
   - Since the loop is now drawn at the very outside, it naturally packs in all the current I flowing through the cable, so ΣI_enc = I.
   - The magnetic field formula is the classic formula outside an infinitely long straight wire: $B = \frac{\mu_0 I}{2\pi r}$.

Exercise 10-10

Answer

Given a coaxial cable consisting of an inner conductor cylinder (radius a) and an outer conductor tube (inner radius b, outer radius c). The two parts carry currents of the same magnitude I but in opposite directions, uniformly distributed over their cross-sections.

By Ampère's circuital law:

$$B \cdot 2\pi r = \mu_0 \sum I_{\text{enclosed}} \implies B = \frac{\mu_0 \sum I_{\text{enclosed}}}{2\pi r}$$

(1) When $r < a$ (inside the conductor cylinder): The loop encloses only part of the current of the inner conductor cylinder. The current density of the inner conductor is $j_1 = \frac{I}{\pi a^2}$, and the enclosed current is:

$$\sum I_{\text{enclosed}} = j_1 \cdot \pi r^2 = I\frac{r^2}{a^2}$$

Hence the magnetic induction in this region is:

$$B = \frac{\mu_0 I r}{2\pi a^2}$$

(2) When $a < r < b$ (between the two conductors): The loop encloses the entire current I of the inner conductor, but not yet the current of the outer conductor.

$$\sum I_{\text{enclosed}} = I$$

Hence the magnetic induction in this region is:

$$B = \frac{\mu_0 I}{2\pi r}$$

(3) When $b < r < c$ (inside the outer conductor): The loop completely encloses the current I of the inner conductor (taken as positive), and also encloses part of the current of the outer conductor (whose current is reversed, taken as negative). The cross-sectional area of the outer conductor is $\pi(c^2 - b^2)$, and the magnitude of the outer-conductor current enclosed by the loop is:

$$I_{\text{enc,outer}} = I\frac{r^2 - b^2}{c^2 - b^2}$$

Hence the net current is:

$$\sum I_{\text{enclosed}} = I - I_{\text{enc,outer}} = I\left(1 - \frac{r^2 - b^2}{c^2 - b^2}\right) = I\frac{c^2 - r^2}{c^2 - b^2}$$

Hence the magnetic induction in this region is:

$$B = \frac{\mu_0 I}{2\pi r}\frac{c^2 - r^2}{c^2 - b^2}$$

(4) When $r > c$ (outside the cable): The loop encloses all the currents of the inner and outer conductors. Since the two are equal in magnitude and opposite in direction, they cancel completely.

$$\sum I_{\text{enclosed}} = I - I = 0$$

Hence the magnetic induction in this region is:

$$B = 0$$

Solution walkthrough

This is a very classic physics model of the coaxial cable; it mainly tests the flexible application of Ampère's circuital law in four different regions.

1. **Region 1: inside the innermost solid cylinder ($r < a$)**
    - Here the current is uniformly distributed over the solid disk of radius a.
    - The Gaussian loop of radius r encloses only the fraction of the current given by the area ratio $\frac{r^2}{a^2}$.
    - The enclosed current is $I\frac{r^2}{a^2}$. Substituting into the formula gives $B = \frac{\mu_0 I r}{2\pi a^2}$ (the field grows linearly with the radius).
2. **Region 2: the gap between the two conductors ($a < r < b$)**
    - In this gap, our Ampèrian loop wraps around the entire inner core (radius a), so the current it encloses is the full I.
    - The outer tube has not been touched yet, so it needs no consideration.
    - Applying the formula directly gives the classic field outside a straight wire: $B = \frac{\mu_0 I}{2\pi r}$.
3. **Region 3: within the metal thickness of the outer conductor tube ($b < r < c$)**
    - This region is more complicated; the loop encloses two things:
        1. The entire innermost core, with current $+I$ (assume upward).
        2. The layer of the outer tube from its inner wall b to the current position r, which carries the reversed current (downward).

- The enclosed area of the outer tube is $\pi(r^2 - b^2)$ and its total cross-sectional area is $\pi(c^2 - b^2)$. By proportion, the magnitude of the enclosed reversed current is $I\frac{r^2-b^2}{c^2-b^2}$.
- Canceling the forward current against the reversed current gives the net current inside the loop: $I - I\frac{r^2-b^2}{c^2-b^2} = I\frac{c^2-r^2}{c^2-b^2}$.
- Substituting into the formula gives $B = \frac{\mu_0 I}{2\pi r}\frac{c^2-r^2}{c^2-b^2}$.

4. **Region 4: the very outside of the cable ($r > c$)**
   - Here comes the most magical part — it is also the principle behind the interference immunity of coaxial cables.
   - Our loop encloses the whole cable (inner core and outer tube).
   - However much current flows out through the core, the same flows back through the outer tube: the two currents are equal and opposite, so the net macroscopic charge flow is 0 (ΣI_enc = I − I = 0).
   - Therefore, the magnetic field outside the coaxial cable is $B = 0$. In other words, the cable produces no magnetic interference to the outside world, and the outside world cannot easily interfere with it either!